

Pole Generators (Straight and Tapered/RTP)

What is it:

- A customizable assembly file which uses I-Logic Rules (Code within Inventor) and Forms to automate the 3D modeling of “special” straight and tapered poles.

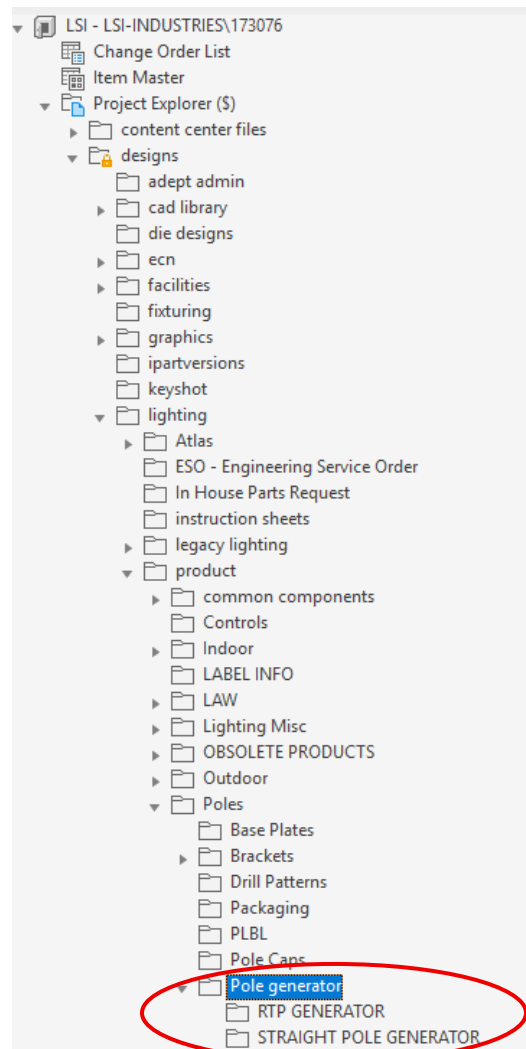
Why:

- Instead of manually modeling unique pole configurations for every order, designers use a single master assembly and a Form UI.

How To Use:

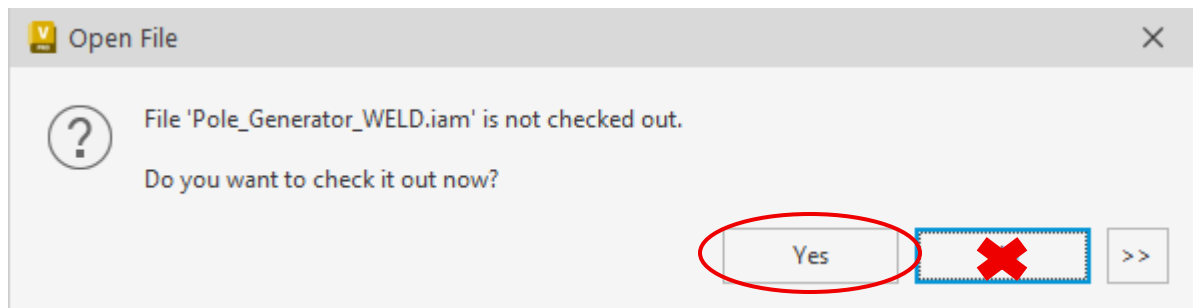
Step1: Find and Check Out/Open Needed Generator

- Both Straight and tapered (RTP) generators are located in Vault
 - designs/lighting/product/Poles/Pole generator



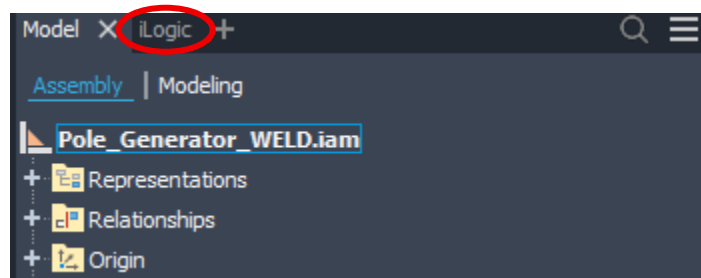
- The two generators are separated into folders as well as their corresponding independent parts.

- **Must “Check Out” the weldment assembly.**

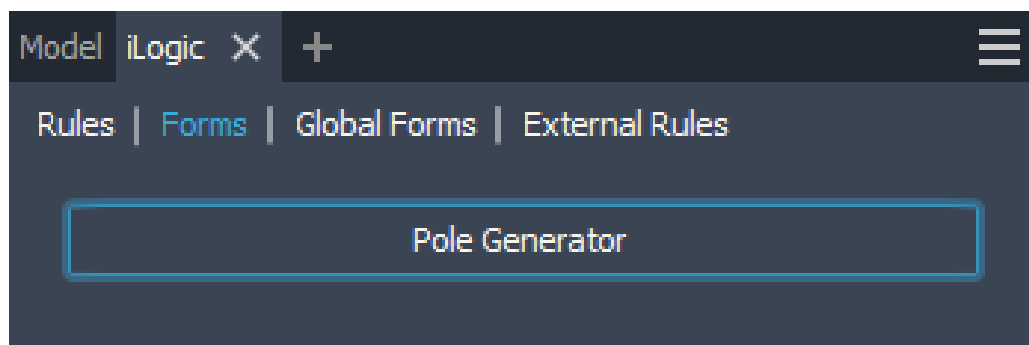


Step 2: I-Logic Form

- First, make sure you have the **I-Logic tab** above the **model browser**
 - If not, press the **plus sign** and select “**I-Logic**” (Not “I-Logic Log”)



- Under the **I-Logic Tab**, you'll find four categories: Rules, Forms, Global Forms, and External Rules.
- Click on “**Forms**” to find a box labeled “**Pole Generator**”. Click this to open the form.



- The form contains **three tabs** separating the pole, base plate, and others.

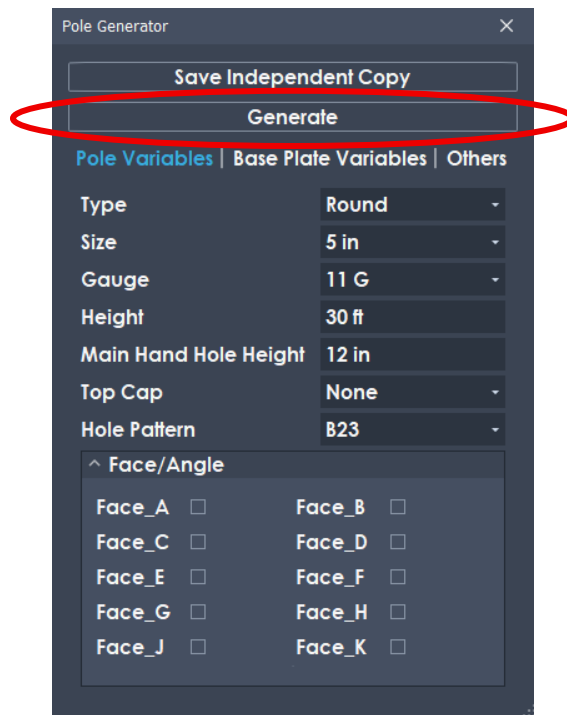
The image displays three sequential screenshots of the 'Pole Generator' software interface, each showing a different tab selected in the top navigation bar.

- First Screenshot (Pole Variables tab):** Shows settings for the pole itself. Variables include Type (Round), Size (5 in), Gauge (11 G), Height (30 ft), Main Hand Hole Height (12 in), Top Cap (None), and Hole Pattern (B23). There is also a section for Face/Angle with checkboxes for Face_A through Face_K.
- Second Screenshot (Base Plate Variables tab):** Shows settings for the base plate. Variables include Thickness (3 in / 4 ul), Minimum Bolt Circle (8 in), Maximum Bolt Circle (11 in), Bolt Size (0.75 in), Square Size (10.125 in), and Slot to Edge (0.735912703473988 in).
- Third Screenshot (Others tab):** Shows settings for additional features. It includes expandable sections for Extra Hand Holes, Coupling Holes, and Tenon.

- Each group under “Others” contains a “**Reset**” button. When pressed, all variables of that group will be set to 0, suppressing them.

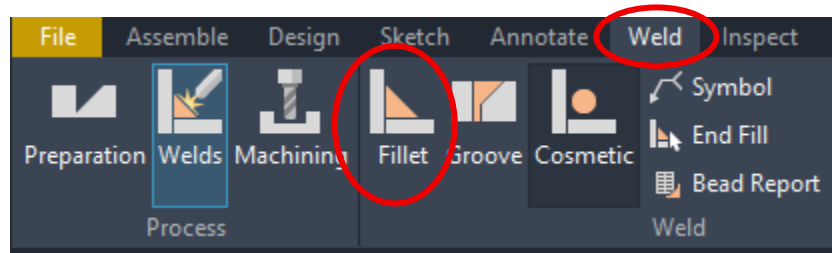
This screenshot shows the 'Pole Generator' interface with the 'Others' tab selected. The 'Extra Hand Holes' group is expanded, showing a list of faces (Face A through Face K). A red circle highlights the 'Reset' button located at the bottom of this group, indicating its function to reset all variables in the group to 0.

- Once you have your variables and features set to your specifications, **you MUST PRESS “GENERATE”** at the top, regardless of some features possibly already being suppressed/unsuppressed.



Step 3: Weldments

- Once generated, you will have to **manually** add 0.25” fillet welds to whatever is required.

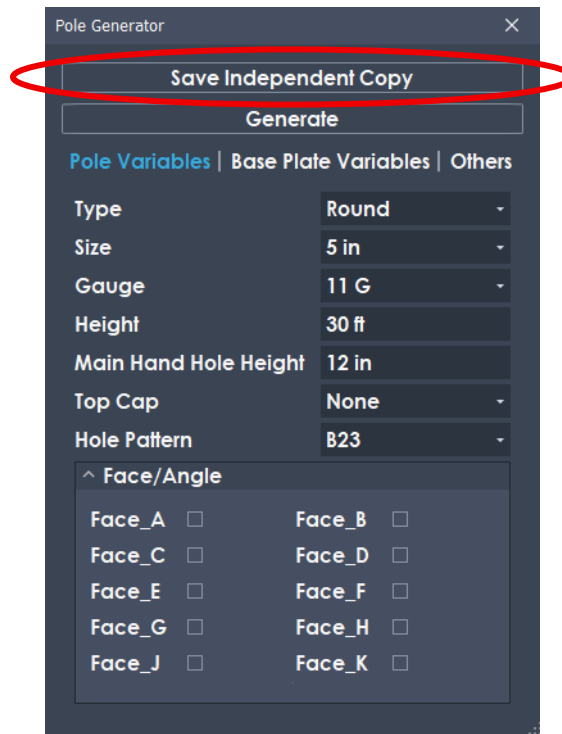


Step 4: Save an Independent Copy of your pole/generator

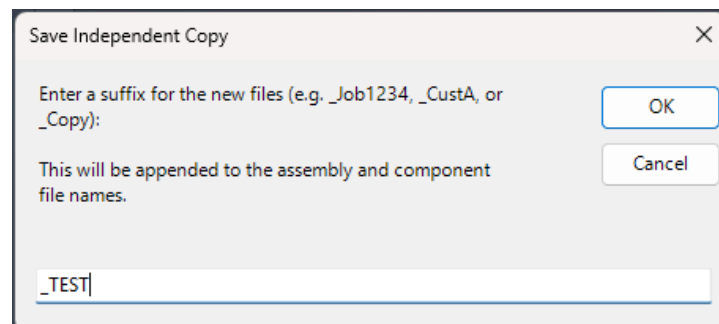
This process can be done before or after customizing your pole as your copy will have all the exact same capabilities

- A button labeled **“Save Independent Copy”** will always be at the top of your form with the “Generate” button and your tabs.

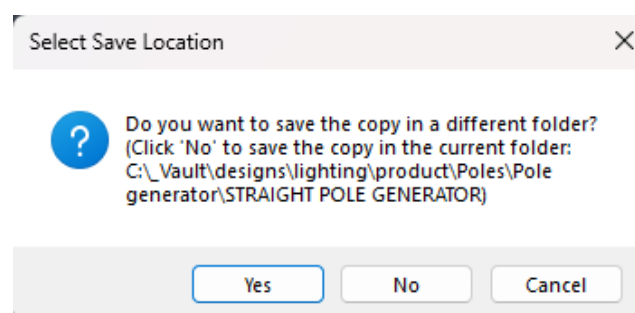
- This button starts the process of **creating a copy** of all components in use, **cutting links/couples** from the main generator, and creating a folder to keep your generator and corresponding part files together.



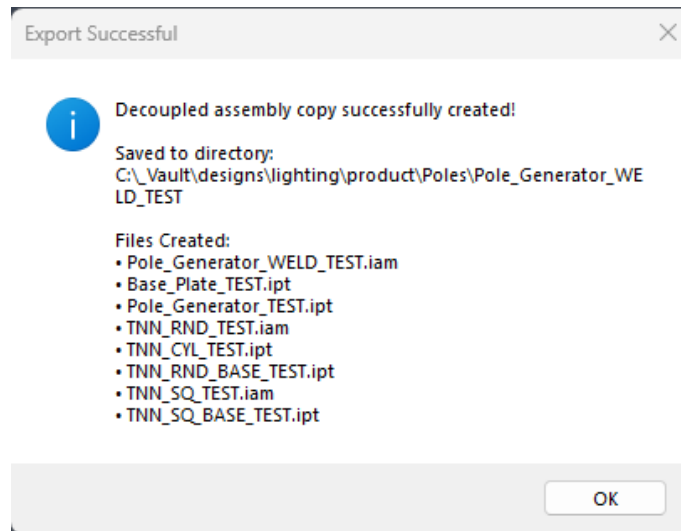
- Once pressed, you'll be prompted to **add a suffix** for your new file.



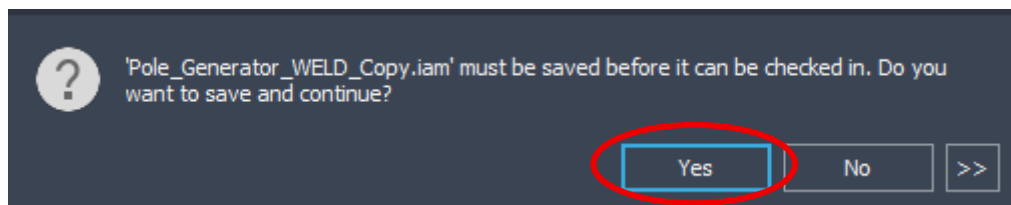
- Next, you'll be asked if you want to **save the copy to a different folder**.
 - If “No” is selected, the copy will be saved in the same file as the main generators.
 - If “Yes” is selected, you have the option to save it anywhere, **as long as it stays withing the “_Vault” project folder/scope**.



- You'll then see a **confirmation box**, displaying where your folder is located and all the files that are held within it.



- The next question asks if you'd like to **check the folder into Vault**.
 - If “No” is selected, the folder will be saved locally in whatever scope/folder you chose but **WILL NOT** be checked into Vault.
 - If “Yes” is selected, the copy will be opened in Inventor, and you'll be asked to save the assembly file. **The folder/files will only be checked into Vault if you “save and continue”.**



- Now, within your independent copy, you have the ability to do whatever you want, including placing more parts such as the actual couplings or editing a past design quickly instead of having to restart with the main generator.
- **REMEMBER TO CHECK MAIN GENERATORS AND COMPONENTS BACK IN FOR OTHERS TO USE**

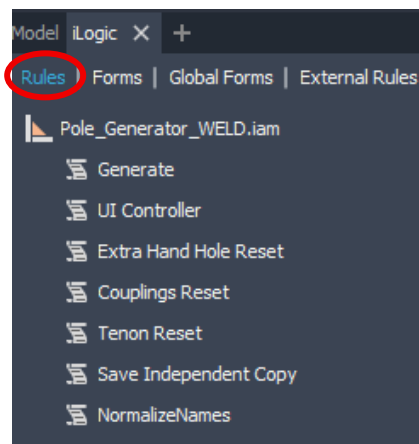
How it Works:

Step 1: Input & Validation

- **UI Input:** The user inputs pole parameters (Height, Gauge, Baseplate Size) directly into the "Pole Generator" form.
- **Rule Trigger:** Parameter changes automatically trigger the "UI Controller" rule to run. This cleans parameter units and resets any unrecognized inputs to template defaults. This helps other rules run smoothly and ensures geometry can't break easily.

Step 2: The Rules (Code)

These rules validate user inputs, dynamically update the 3D pole geometry, and export production-ready files.



- Both Generators have 7 Rules that work together with the form.
 - **"Generate"**: Acts as the builder; it pushes parameter values down to the parts, toggles 3D features (cuts, handholes, tenons, and caps) on or off, and rebuilds the physical model.
 - **"UI Controller"**: Acts as the gatekeeper; it validates user inputs in the form, enforces physical limits (like max height and baseplate size), and calculates slot widths all before the model rebuilds.
 - **"Extra Hand Hole Reset"**: Resets all extra handhole offsets to zero to suppress additional hole cutouts.
 - **"Couplings Reset"**: Resets all coupling heights to zero to clear all coupling cuts from the pole faces.
 - **"Tenon Reset"**: Instantly resets all tenon parameters to zero to suppress the tenon attachment.
 - **"Save Independent Copy"**: Publishing utility that clones the entire assembly and its parts into a separate folder, updates their internal references, and creates a standalone copy of the job.

- **“NormalizeNames”**: Clean-up utility that renames components in the assembly tree back to their defaults, ensuring the other rules (mainly “Save Independent Copy”) don't lose track of parts due to renaming.
- Main rules/limits breakdown
 - **Shared Rules & Limits (Both Generators)**
 - **Baseplate Clearance**: Baseplate size must be at least 2 inches larger than the pole size.
 - **Slot Sizing**: Baseplate slot width is locked to: Bolt Size + 0.125 in.
 - **Edge Safety Pre-Check**: Prevents model crash by stopping the rebuild and reverting to last valid configuration if slot-to-edge distance falls below 0.175 inches.
 - **Nut Clearance**: Calculates minimum bolt circle to clear nut corners based on bolt size, automatically resetting the bolt circle if it is set too small.
 - **Bolt Circle Bounds**: The maximum bolt circle cannot be set smaller than the minimum clearance bolt circle.
 - **Tenon Proportions**: Tenon base must be at least 0.5 inches larger than top pole OD. Tenon OD must be at least 1.75 inches and at least 0.5 inches smaller than the tenon base size.
 - **Accessory Suppression**: Clamps coupling and extra handhole heights below overall pole height to prevent floating cuts.
 - **Face Conflicts**: Automatically disables Face K/H and Face G/J combinations if they are checked together.
 - **Straight-Only Rules & Limits**
 - **Pole Shapes**: Supports both Square and Round shapes.
 - **Sizes & Gauges**:
 - Square poles: Available in 4", 5", 6" sizes; 11 G and 07 G gauges (10 G disabled).
 - Round 4" poles: Locked to 10 G gauge.
 - Round 5" poles: Available in 11 G and 07 G gauges (10 G disabled; 6" size disabled).
 - **Handhole Reinforcement**: Swaps standard handholes for reinforced ones dynamically based on height:
 - Square poles: Reinforced at 22 feet or taller.
 - Round poles: Reinforced at 18 feet or taller.
 - **Tenon Styles**: Supports both Square and Round tenon subassemblies.
 - **Top Cap Selection**: Standard/None cap parts are toggled dynamically in the assembly by size/shape. Top cap is overridden to "None" if a tenon is active.
 - **Face View**: Hides faces E through K in the UI form when "Square" is selected.
 - **Round Tapered-Only Rules & Limits**
 - **Pole Shapes**: Supports Round Tapered shapes only.
 - **Sizes & Gauges**: Available in 5.8", 6.6", 7.5", 8.5", 9" sizes; 11 G and 07 G gauges only.
 - **Overall Height Limits**: Enforces maximum pole heights based on size:
 - 5.8" Size: Max 20 feet.
 - 6.6" Size: Max 25 feet.

- 7.5" Size: Max 30 feet.
- 8.5" Size: Max 35 feet.
- 9.0" Size: Max 39 feet.
- **Accessory Height Limits:** Enforces size-specific maximum height limits for extra handholes (XHH):
 - 5.8" Size: Max 13 feet.
 - 6.6" Size: Max 19 feet.
 - 7.5" Size: Max 25 feet.
 - 8.5" Size: Max 32 feet.
 - 9.0" Size: Max 36 feet.
- **Handholes:** Always uses reinforced handholes and backing frames (no unreinforced option).
- **Tenons:** Supports Round tenons only. Does not manage Top Cap assembly components in the rules.

Step 3: Model Rebuild (CAD)

- **Name Normalization:** Renames active assembly components back to standard templates to prevent broken parameter links.
- **Parameter Push:** Transfers validated assembly-level values directly down to nested parts and sketches (Pole, Baseplate, Tenon).
- **Model Update:** Activates or suppresses 3D CAD features (cutouts, reinforcements, subassemblies) and rebuilds the physical model geometry.

Step 4: Export

- **User Setup:** Prompts the user to specify a target directory and a filename suffix (e.g. _Job1234).
- **File Cloning:** Copies the custom assembly and part files to a new separate folder
- **Reference Re-linking:** Invisibly opens the new assemblies, re-links internal file paths to point to the new copies, and prompts for a Vault check-in.

Troubleshooting:

99% of errors will be found within the rules/code and the sketches (broken geometry).

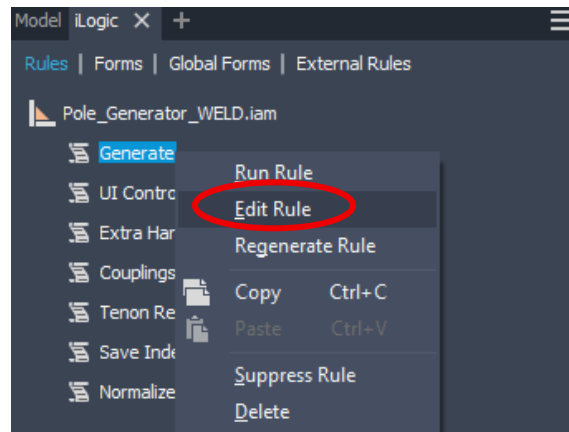
If something simply doesn't suppress or unsuppress as it should, try pressing the "Generate" button again.

HIGHLY SUGGEST USING A.I. TO ASSIST YOU WITH ANY CHANGES BEING DONE TO THE CODE

1. Rule Errors

- You should never have to touch any rules other than the main two: **"Generate"** and **"UI Controller"**.

- To edit/look at the code, make sure you are under the **I-Logic Rules tab**, right click the rule u want to see, and select “**Edit Rule**” .



- At the top of our main rules (“Generate” and “UI Controller”) is an **index** which breaks down the different sections of code and explains them.
 - Use this to pin point where an error is exactly or where you’d like to change something.

```

1 | =====
2 | INDEX / TABLE OF CONTENTS (Copy/paste "[SEC-#]" into Ctrl+F to jump)
3 | =====
4 | [SEC-01] File Identifier - Dynamically maps active component names by file path/extension (Helps with creating independent copies of the generator)
5 | [SEC-02] Occurrence Check 1: Face Conflicts (H/K) - Prevents Face K and H from being selected together
6 | [SEC-03] Occurrence Check 2: Face Conflicts (G/J) - Prevents Face G and J from being selected together
7 | [SEC-04] Occurrence Check 3: Nut Clearance (Bolt Circle) - Calculates minimum bolt circle to clear corners
8 | [SEC-05] Occurrence Check 4: Tenon Sizing Limits - Enforces tenon base size minimum and max/min limits of tenon OD
9 | [SEC-06] Occurrence Check 5: Accessory Height Clamping - Clamps handhole and coupling heights below overall pole height
10 | [SEC-07] Occurrence Check 6: Baseplate Clearance - Ensures Baseplate Size is at least 2 inches larger than Pole Size
11 | [SEC-08] Occurrence Check 7: Slot-to-Edge Pre-Check - Mathematical pre-validation check of Slot-to-Edge distance
12 | [SEC-09] Locate Assembly Components - Finds the Pole part definition and resets/suppresses cut features
13 | [SEC-10] Linking Parameters - Pushes assembly parameter values down to the nested parts
14 | [SEC-11] Square Pole Configurations - Configures Square pole specific features
15 |     [SEC-11a] Square Baseplate Hole Profile - Activates square center hole cutout in the baseplate
16 |     [SEC-11b] Square Handhole Reinforcement - Toggles hand hole reinforcements based on height
17 |     [SEC-11c] Square Top Cap Selection - Activates matching size cap component for Square poles
18 | [SEC-12] Round Pole Configurations - Configures Round pole specific features
19 |     [SEC-12a] Round Baseplate Hole Profile - Activates Round center hole cutout in the baseplate
20 |     [SEC-12b] Round Handhole Reinforcement - Toggles frame reinforcements based on height
21 |     [SEC-12c] Round Top Cap Selection - Activates matching size cap component for Round poles
22 | [SEC-13] Tenon Weldment Activation - Controls Square/Round Tenon subassemblies and nested parameters
23 | [SEC-14] Pole Gauge & Thickness Variables - Maps sheet metal Gauge to thickness and radius
24 | [SEC-15] Hole Pattern Activation - Renders lighting fixture cutouts (B2/B3/B5/B23) on faces A-K
25 | [SEC-16] Extra Handholes & Box Activation - Manages extra handholes and reinforcing frames
26 | [SEC-17] Coupling Activation - Activates coupling cuts on faces A-K based on height parameters
27 | [SEC-18] Final Rebuild & Bolt Circle Validation - Rebuilds 3D model and validates bolt circles
28 | =====

```

- Next to every line is a small note explaining what its saying/doing.

2. Sketches (Geometry breaks)

- The geometry of sketches break when specific variables are input, that are **mathematically and/or physically impossible**.
 - Occurrence checks and limits have been put in place to avoid this as much as possible.
- If this happens, you’ll need to investigate the sketch of whatever part or feature is affected

- Most common errors are a tangent constraint flipping, causing your extrusion feature to disappear, or a missing constraint in general.